

Q1. Create a dataframe containing information about different products such as Price, Rating, and Number of Sales. Generate a Dendrogram to group similar products based on their characteristics.

[10 Marks]

Q2. Using the Car Price Prediction dataset, apply Polynomial Regression to predict the Selling Price based on Vehicle Age, Present Price, and Kilometers Driven. Evaluate the model using the R^2 Score and visualize the actual data points along with the fitted polynomial curve.

[20 Marks]

OR

Q2. Create two DataFrames containing information about different mobile phones.

DataFrame 1: Brand, Model, Price, RAM

DataFrame 2: Model, Storage, Battery_Capacity, Operating_System

Perform the following data transformation techniques:

- a) Data Integration by combining the two DataFrames using Model
- b) Label Encoding on the Brand attribute of the merged DataFrame
- c) One-Hot Encoding on the Operating_System attribute of the merged DataFrame

Display the original DataFrames, merged DataFrame, and transformed DataFrame.

[20 Marks]

Q3. Viva

[5 Marks]